

Management of Impetigo

Questions	Answer here	For pharmacist
Age: Is the patient aged ≥ 2 years of age?	Age years	[Triggers: < 2 years $\rightarrow$ Refer to GP]
Consent: Does the patient/carer/caregiver consent? (*The patient must be physically present in the pharmacy)	Yes / No	[Triggers: No $\rightarrow$ Refer to GP]
Symptoms: Does the patient present with signs, symptoms, or a history consistent with impetigo, based on initial presentation and information provided to the pharmacist?	No / Yes Examples of likely impetigo: - Golden or honey-coloured crusted lesions - Lesions commonly on the face, limbs, or around the mouth or nose Vesicles or pustules that rupture easily - May be mildly itchy or red, but not deep, painful, or scaly Erythema may be present where the lesions or sores are located	[Triggers: No $\rightarrow$ Consider other skin conditions (e.g., shingles, psoriasis, acne, atopic dermatitis, HSV, scabies) $\rightarrow$ Provide Pharmacist Care and/or Refer to GP]
Risks: Does the patient report/present with any of the following?	No / Yes - The patient is identified with or at risk of recurrent impetigo (e.g., symptoms have not resolved after the first course of antibiotic treatment, symptoms significantly or rapidly worsen, if impetigo infection reoccurs frequently) - The patient is immunocompromised - The patient is at high risk of complications of impetigo, including patients at high risk of ARF (see table below)	[Triggers: Yes $\rightarrow$ Refer to GP]
Red flag symptoms: Does the patient report/present with any of the following red flag criteria?	No / Yes Widespread, painful rash that may be erythematous Non-blanching purple rash - Blistering of the skin and/or mucous membranes (that may include mouth and eyes) - Signs and symptoms of serious and/or systemic illness (e.g., complicated cellulitis, severe ecthyma, acute ARF) including fever, lethargy, headache, rash, nausea and vomiting, severe pain, sore and swollen joints, food/drinking aversion in a child - Generalised erythema that covers 90% or more of the skin surface, especially when associated with systemic symptoms	[Triggers: Yes $\rightarrow$ Refer to the Emergency Department immediately]
Severe symptoms & differential Dx: Does the patient report/present with any of the following?	No / Yes - Signs of bullous impetigo (large, flaccid blisters) - Signs that impetigo is widespread, severe and/or has ecthyma (ulceration, induration) present Chronic sores or ulcers - The patient presents with generalised erythema that covers 90% or more of the skin surface, but the patient is otherwise well, with no systemic features. - A clear diagnosis of impetigo cannot be made (i.e. uncertain diagnosis of uncomplicated non-bullous impetigo), and/or other/co-occurring secondary conditions are suspected that cannot be treated in the community pharmacy setting, e.g., Infected atopic or discoid eczema, herpes simplex virus (HSV-1), varicella (chickenpox), herpes zoster (shingles), cellulitis, scabies, psoriasis, folliculitis or acne, contact dermatitis, dermatophytosis (tinea), candidiasis, thermal burns, and molluscum contagiosum.	[Triggers: Yes $\rightarrow$ Refer to GP]
Non-bullous impetigo: Does the patient present with a clear presentation of non-bullous impetigo?	No / Yes Key Features of non-bullous impetigo (eligible for pharmacist treatment) Honey-coloured crusts, typically on face/extremities Vesicles/pustules that rupture easily - Mild itch, minimal pain - No systemic symptoms	[Triggers: No $\rightarrow$ Refer to GP] ***Note: Pharmacists should treat only clearly eligible low-risk cases and refer to a GP for swabbing or further review if the diagnosis is uncertain, recurrent, or not improving.
Number of lesions: Are the number of lesions/sores confirmed by the pharmacist as limited and not extensive (≤ 2 sores)? If there are > 2 sores/lesions, how many body regions are they confined to?	Number of lesions/sores lesions How many body regions involved Body regions If (≤ 2 sores $\rightarrow$ check other risks before proceed to pharmacist care) (> 2 sores / 1 body region $\rightarrow$ check other risks before proceed to pharmacist care) (> 2 sores / 2 body regions $\rightarrow$ check other risks before proceed to pharmacist care + Refer to GP for follow up.)	[Triggers: ($\bullet > 2$ sores / > 2 body regions $\rightarrow$ Refer to GP)]

*** Patients with impetigo do not require an initial skin swab before starting empirical therapy unless the diagnosis is uncertain or impetigo is recurrent. A skin swab for culture and susceptibility testing is recommended for patients with impetigo that is not responding to appropriate empirical therapy.

Table 2: People considered at high risk of developing ARF

Individuals at high risk of developing ARF	
- Aboriginal and Torres Strait Islander people residing in a rural or remote area, or living in a household affected by household overcrowding (> 2 people per bedroom) or experiencing socioeconomic disadvantage. - Māori and/or Pacific Islander person living in a household affected by overcrowding (> 2 people per bedroom) or experiencing socioeconomic disadvantage. - A person with a personal history of ARF or RHD. - A person with a family or household member with a recent history of ARF or RHD.	
Additional risk factors for individuals aged ≤ 40 years (particularly between 5-20 years)	
- People living in a household affected by household overcrowding (> 2 people per bedroom) or experiencing socioeconomic disadvantage. - People with current or prior residence in (or frequent or recent travel to) an area with a high rate of ARF e.g., refugees and migrants from low-middle income countries, rural and remote communities.	

Empirical therapy for impetigo for adults

Patients with non-recurrent localised non-bullous impetigo (≤ 2 sores, non-recurrent, not at high risk of ARF)			
	• Topical ointment or cream, mupirocin 2%	Apply to lesions, every 8 hours for 5 days (2%TDSx5days)	Note: *** Mupirocin is first line ; hydrogen peroxide is second line
	• Topical cream, hydrogen peroxide 1%	Apply to lesions, every 8 hours for 5 days (1%TDSx5days)	
Patients with extended (non-recurrent, limited to no more than 2 sites, not at high risk of ARF), recurrent or bullous impetigo .			
	• Oral, dicloxacillin or flucloxacillin	500 mg, every 6 hours for 5 days* (500QIDx5days)	Note: *** Recommended first-line oral therapy ***Stop therapy after 3 days if infection has resolved
	• Oral, cefalexin	1,000 mg every 12 hours for 5 days* (1000BDx5days)	Note: *** First-line therapy – can be used if the patient has had a non-severe (immediate or delayed) penicillin hypersensitivity *** Less frequent dosing and may be preferred by patients ***Stop therapy after 3 days if infection has resolved
	• Oral, trimethoprim + sulfamethoxazole	160 + 800 mg , every 12 hours for 3 days* OR 320 + 1,600 mg , daily for 5 days*	Note: ***Second-line therapy – use if the patient has had a severe (immediate or delayed) penicillin hypersensitivity *** Once-daily regimen may be preferred for administration in school settings
	• Intramuscular, benzathine benzylpenicillin (benzathine penicillin G)	1.2 million units (2.3 mL) , as a single dose	Note: *** Long acting – 4 weeks *** Caution, do not confuse with the short acting benzylpenicillin *** Hydrolyses to benzylpenicillin

*Consult specialised references for further information about dosing in renal impairment.

Empirical therapy for impetigo for children

Patients with non-recurrent localised non-bullous impetigo (up to 2 sores) who are not at high risk of ARF			
	• Topical ointment or cream, mupirocin 2%	Apply to lesions, every 8 hours for 5 days (2%TDSx5days)	Note: *** Mupirocin is first line ; hydrogen peroxide is second line
	• Topical cream, hydrogen peroxide 1%	Apply to lesions, every 8 hours for 5 days (1%TDSx5days)	
Patients with extended (non-recurrent, limited to no more than 2 sites, not at high risk of ARF), recurrent or bullous impetigo .			
	• Oral, cefalexin	25 mg/kg up to 1,000 mg every 12 hours for 5 days*	Note: *** Children tolerate cefalexin liquid better than dicloxacillin or flucloxacillin due to taste *** can be used if the patient has had a non-severe (immediate or delayed) penicillin hypersensitivity ***Stop therapy after 3 days if infection has resolved
	• Oral, dicloxacillin or flucloxacillin	12.5 mg/kg up to 500 mg, every 6 hours for 5 days*	Note: ***Stop therapy <u>after 3 days</u> if infection has resolved
	• Oral, trimethoprim + sulfamethoxazole (1 month+)	≥1 month old: 4 + 20 mg/kg up to 160 + 800 mg, every 12 hours for 3 days* OR ≥1 month old: 8 + 40 mg/kg up to 320 + 1,600 mg, daily for 5 days*	Note: ***Second-line therapy – use if the patient has had a severe (immediate or delayed) penicillin hypersensitivity ***Once-daily regimen may be preferred for administration in school settings
	• Intramuscular, benzathine benzylpenicillin (benzathine penicillin G)	Dose by weight: ≥20 kg: 1.2 million units (2.3 mL) , as a single dose 10–<20 kg: 0.6 million units (1.2 mL) , as a single dose <10 kg: 0.45 million units (0.9 mL) , as a single dose	Note: ***Long acting – 4 weeks ***Caution, do not confuse with the short acting benzylpenicillin ***Hydrolyses to benzylpenicillin